2. Credit Derivatives

Defining Structured Products

The traditional role of *banks* is to take deposits and to make loans. The interest charged on loans is greater than the interest paid on deposits. The difference between the two covers administrative costs and loan losses while providing a satisfactory return on equity. Banks offer lines of credit to businesses and individual customers. This exposes them to several risks. Central bank regulators require banks to hold capital for the risks they bear. The objective of regulators is to keep the total capital of a bank sufficiently high that the chance of a bank failure is very low.

Credit risk is the risk that counterparties in loan transactions and derivative transactions will default. This has traditionally been the greatest risk facing a bank and is usually the one for which the most regulatory capital is required.

Consider a *mutual fund* (for stocks). Mutual funds invest money on behalf of individuals and companies. The funds from different investors are pooled and investments are chosen by the fund manager in an attempt to meet specified objectives. One of the attractions of mutual funds is the diversification opportunities they offer. Diversification improves an investor's risk-return trade-off.

Now consider pooling different loans (bank's assets) together to create a new financial product to offer to investors. If we divided these pooled assets equally, there would be a mismatch between the riskiness of these securities and the risk preference/ levels of risk-tolerance of available investors. There is therefore a demand for products that offer unique risk-return profiles that would match the risk appetite of these investors.

First, all of the loans originated by the bank would need to "move off" their balance sheet. The bank therefore "sells" these "assets". The funds that they receive gives them more cash (capital) which allows them to make more loans. This process also transfers the risk of default from the bank to another party.

Securitization, is where the originators of mortgages sold portfolios of mortgages to companies that created products for investors from them. Securitization is an important and useful tool for transferring risk in financial markets. It provides liquidity, facilitates the transfer of risk, and presents unique risk-return opportunities to buyers of such securities.

Structuring is the process of engineering unique financial opportunities from existing asset exposures. Structured products are asset-backed securities that apply the concept of structuring to cash flows from a portfolio of debt securities into multiple claims. These claims are referred to as tranches. A tranche is a distinct claim on assets that differs substantially from other claims in aspects such as seniority, risk, and maturity. In this section we will cover two **types of Structured Products: Credit Default Swaps (CDS) and Collateralized Debt Obligations (CDOs).**

Credit risk emanates from the structuring of cash flows: Cash flows are promised but are backed by an uncertain ability to meet those contractual obligations. The structuring of cash flows is the process of dividing a stream of cash into two or more streams that differ about timing, risk, taxation, or other attributes. The simplest and most common structuring of cash flows is that of the traditional corporate structure of an operating firm that divides the cash flows from the firm's assets into a fixed stream paid to debt holders and a residual stream available to equity holders. The fixed commitments such as bonds that result from structuring usually contain credit risk.

Types of Credit Risk

Credit risk is dispersion in financial outcomes associated with the failure or potential failure of a company or counterparty to fulfil its obligations. Credit risk is a measure of the creditworthiness of a borrower. In calculating credit risk, lenders are gauging the likelihood they will recover all of their principal and interest when making a loan. Borrowers considered to be a low credit risk are charged lower interest rates. Credit risk, unlike market risk, contains substantial idiosyncratic risk and is typically related to the non-performance of a single firm or counterparty. Also, unlike market risk, which may lead to symmetrical payoff distributions, credit risk generally leads to payoff distributions that are substantially skewed to the left. The upside performance of a traditional position exposed to credit risk is limited to the recovery of the original investment plus the promised yield, while the downside performance could lead to the loss of the entire investment.

There are three important types of credit risk:

1. Default risk
2. Downgrade risk
3. Credit spread risk

Default risk is the risk that the issuer of a bond or the debtor on a loan will not repay the interest and principal payments of the outstanding debt in full. A debtor is deemed to be in default when it fails to make a scheduled payment on its outstanding obligations. Default risk can be complete, in that no amount of the bond or loan will be repaid, or it can be partial, in that some portion of the original debt will be recovered.

Downgrade risk is the risk that the creditworthiness of a borrower has changed, as indicated by the decision of a national rating agency to change the borrower's credit rating. The credit rating agencies call this *ratings migration.*

Ratings migration is a positive or negative change in a credit rating. An increase in credit rating indicates a perceived and/or real improvement in the financial health of a company, and investors normally react by bidding up the price of the company's related debt (lowering the yield and the yield spread). Conversely, a downward ratings migration indicates a perceived and/or real deteriorating financial condition and results in lower bond prices and higher yields.

Credit spread risk is the risk that the credit spread will change. A credit spread is the amount by which the yield of a risky bond exceeds the yield on a riskless bond of similar maturity.

Credit Derivative Markets

The risk of fluctuations in the value of assets in an economy is borne by both equity holders and debt holders. The portion of the risk borne by equity holders is often termed *market risk*, and the portion held by bond holders is termed *credit risk*. Derivatives are cost-effective vehicles for the transfer of risk. *Credit derivatives* transfer credit risk from one party to another such that both parties view themselves as having an improved position as a result of the derivative. The primary way that credit derivatives contribute to the economy and its participants is by facilitating diversification. Credit derivatives can provide liquidity to the market in times of credit stress. The availability and use of credit derivatives has increased in recent decades, with the result that credit risk has gradually changed from an illiquid risk that was not considered suitable for trading to a risk that can be traded like other sources of risk (e.g., equity, interest rates, and currencies).

Consider the challenge faced by a major bank that has established a long-term relationship with a client such as a traditional operating firm. The bank provides for several of its client's needs, including payment services and credit. If the firm is very large, the credit risk exposure of the bank to the firm through its loans to the firm may become substantial relative to the size of the bank. However, the bank may wish to be the sole direct creditor of the firm for several reasons. First, the bank may view meeting all of the firm's loan needs as increasing the chances that the bank will remain the firm's sole supplier of other services. Second, the bank may wish to avoid the potential conflicts of interest and legal complexities of making loans to a firm alongside other creditors. As the sole creditor, the bank may be better able to pursue its self-interest. Credit derivatives can provide the bank with a cost-effective solution: The bank can make large loans to the firm and transfer as much risk as the bank desires to other market participants through credit derivatives. At the same time, other banks may similarly transfer their risks to other banks. Through this process, each bank may be able to hold relatively well-diversified portfolios of credit risks while maintaining efficient and effective relationships with key clients. While the banks are reducing their credit risk to their borrowers through credit derivatives, they are increasing their counterparty risk to the banks from whom they purchase credit derivatives.

Three groupings of credit derivatives

- <u>*Single-name versus multiname instruments:*</u> *Single-name* credit derivatives transfer the credit risk associated with a single entity. This is the most common type of credit derivative and is often credit default swaps. Single-name CDS are the most popular way to allow one party to buy credit protection from another party.

  Multiname instruments, in contrast, make payoffs that are contingent on one or more credit events (e.g., defaults) affecting one or more reference entities. An example of this includes a collateralised debt obligation.

- <u>*Funded versus unfunded instruments*</u> : *Unfunded* credit derivatives involve exchanges of payments that are tied to a notional amount, but the notional amount does not exchange hands until a default occurs. An unfunded credit derivative is similar to an interest rate swap in which there is no initial cash purchase of a promise to receive principal, but instead only an agreement to exchange future cash flows. The most common unfunded credit derivative is the CDS. Unfunded instruments expose at least one party to counterparty risk. Counterparty risk is the risk associated with the other party to a financial contract not meeting its obligations. Every derivative trade needs to have a party to take the opposite side. Unfunded instruments can be for a single name or for multiple names instruments.

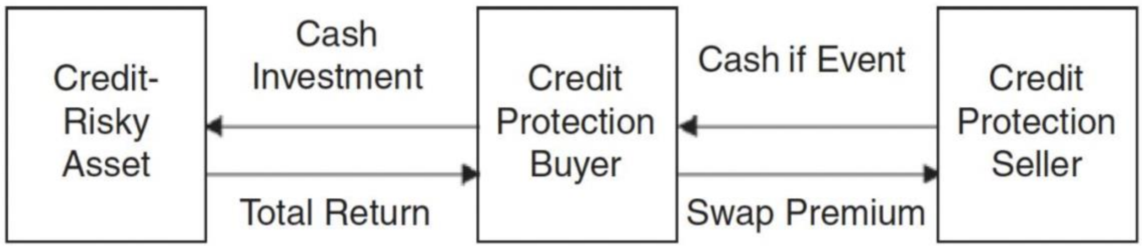
  Funded credit derivatives require cash outlays and create exposures similar to those gained from traditional investing in corporate bonds through the cash market. They can be thought of as a of as a riskless debt instrument with an embedded credit derivative.

- <u>*Sovereign versus nonsovereign entities:*</u>

  The reference entities of credit derivatives can be sovereign nations or corporate entities. Credit derivatives on sovereign nations tend to be more complex because their analysis has to consider not only the inability of the entity to meet its obligations but also the potential unwillingness of the nation to meet its obligations. The modeling of the credit risk associated with sovereign risk involves political and macroeconomic risks that are normally not present in modeling corporate credit risk. Finally, the market for credit derivatives on sovereign nations is smaller than those for other credit derivatives.

Credit Default Swaps

A *credit default swap* is the simplest form of credit insurance. It is simply a bilateral contract where the credit protection buyer pays a periodic fee (analogous to an insurance premium) to the credit protection seller in exchange for a contingent payment if a credit event occurs with respect to an underlying credit-risky asset. A CDS may be negotiated on any of a variety of credit-risky investments—primarily corporate bonds.



Mechanics of a Credit Default Swap

The CDS market is contract-driven. This means each CDS is a privately negotiated transaction between the credit protection buyer and seller. Fortunately, the *ISDA* (International Swaps and Derivatives Association), the primary industry body for derivatives documentation, has established standardized terms for CDS. These terms are not mandated for use but are available to market participants and are used as a framework for negotiating a deal. This section provides some detail regarding the standard ISDA agreement. Standard agreements contain provisions relating to the following five aspects of the deal:

1. *CDS spread*: The premium to be paid by the credit protection buyer is often called the spread, and it is quoted in basis points per annum on the notional value of the CDS. The CDS spreads are not credit yield spreads, but rather price quotes for buying credit insurance. Typically, the price of insurance is paid quarterly by the protection buyer.
2. *Contract size*: ISDA does not impose any limits on size or length of term of a CDS; this is up to the negotiation of the parties involved. Most CDS fall in the range of $20 million to $200 million with a tenor (term) of three to five years.

3. *Trigger events*: This is the heart of every CDS transaction. Trigger events determine when the credit protection seller must make a payment to the credit protection buyer.

Both sides to a CDS negotiate these terms intensely. The broader the definition of a trigger event, the more likely the cash flows from the protection seller to the protection buyer and the higher the appropriate spread. The ISDA provides for six kinds of potential trigger events, but the parties to a CDS can add more, although the six events identified by ISDA cover virtually all types of credit events:

Bankruptcy. A filing for bankruptcy is typically associated with a company's inability to pay its debt.

Failure to pay. While a company may not be in bankruptcy yet, it may not be able to meet its debt obligations as they come due.

Restructuring. This is any form of debt restructuring that is disadvantageous to a holder of the referenced credit. Restructuring is a fuzzy term, and ISDA attempts to clarify this part of the standard contract by offering the following four options for the parties to consider: no restructuring, full restructuring, modified restructuring (which limits obligations to bonds maturing in less than 30 months after a restructuring), and modified restructuring (which is less strict than modified restructuring because bonds can have maturities of up to 60 months after a restructuring).

Obligation acceleration. All bond and loan covenants contain provisions that accelerate the repayment of the loan or bond if the credit quality of the borrower begins to deteriorate due to a number of events, such as a failure to pay, a bankruptcy (which ISDA covers independently), or a ratings downgrade.

Obligation default. This is any failure to meet a condition in the bond or loan covenant that would put the borrower in breach of the covenant. It could be something like the failure to maintain a sufficient current ratio or a minimum interest earnings coverage ratio.

Repudiation/moratorium. This is most frequently associated with sovereign or emerging markets debt. It is simply a refusal by the sovereign government to repay its debt as it comes due or even an outright rejection of its debt obligations.

4. *Settlement*: If a credit event occurs, settlement can be made either with a cash payment or with a physical settlement. In a cash settlement, the credit protection seller makes the credit protection buyer whole by transferring to the buyer an amount of cash based on the contract. The settlement price can sometimes be the present value of the contractual cash flows over its remaining life, or it may be determined through auction processes. Cash settlement does not occur as frequently as one might expect, because it is difficult to agree on a good market-based measure of the loss. Therefore, most CDS use physical settlement upon the occurrence of a credit event. Under physical settlement, the credit protection seller purchases at par value the impaired loan or bond from the credit protection buyer. The credit-risky asset is physically transferred to the credit protection seller's balance sheet, and the face or par value of the bond is transferred to the protection buyer from the protection seller.

5. *Delivery*: Within particular limits, the credit protection buyer has a choice of assets that can be delivered for physical settlement. This raises the issue of which of those assets is cheapest to deliver. The concept of multiple deliverable assets is common throughout derivatives and provides an option to the short option position that should be reflected in the contract's price or terms. Deliverables can include direct obligations of the referenced entity, such as corporate bonds or bank loans, obligations of a subsidiary of the referenced entity if the subsidiary is at least 50% owned by the referenced entity (this is sometimes called qualifying affiliate guarantees), and obligations of a third party that the referenced entity may have guaranteed, known as qualifying guarantees. Note that physical settlement can create problems if there is an insufficient supply of assets to deliver, possibly because a CDS has been overwritten.

Example:

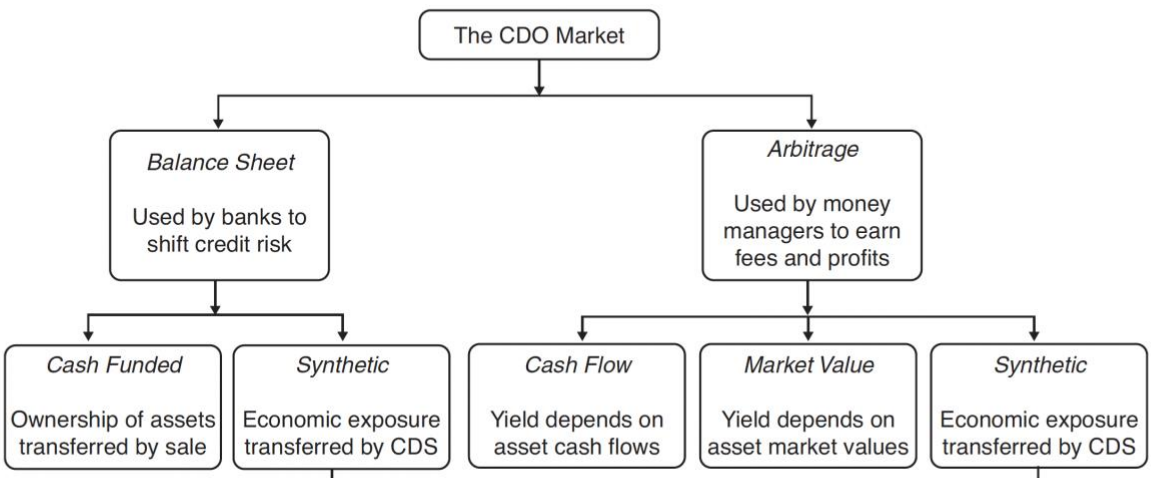
In this example, a hypothetical transaction takes place between a hedge fund (the Fund) as a seller of protection and a commercial bank (the Bank) as a buyer of protection. The reference entity is an airline company (the Firm). The referenced asset is $20 million of face value debt. The term of the transaction is seven years. In exchange for the protection provided over the next seven years, the Fund receives 2% of the notional amount per year, payable quarterly.

The contract will be settled physically. This means that if a credit event takes place, the Bank will deliver $20 million of face value of any qualifying senior unsecured paper issued by the Firm in return for a $20 million payment by the Fund. Further, the contract will be terminated, and no further payments will be made by the Bank.

Let's assume that default takes place after exactly three years. What cash flows and exchanges take place? Each quarter for 12 quarters, the Bank pays the Fund $100,000. This value is found by multiplying the notional amount ($20 million) by the quarterly rate of 0.5% (i.e., 2%/4). When the default occurs, the Bank delivers $20 million in face value of the referenced bond to the Fund in exchange for $20 million in cash. The CDS terminates immediately after these exchanges.

Collateralised Debt Obligations

A collateralized debt obligation (CDO) - applies the concept of structuring to cash flows from a portfolio of debt securities into multiple claims; these claims are securities and are referred to as tranches.



The exhibit below illustrates the concept of a CDO being used to structure the cash flows from a portfolios of high yield bonds. There are $100 million of high-yield bonds on the left-hand-side of the exhibit that serve as the assets or collateral portfolio (or pool) for the structure. These bonds generate cash flows in the form of coupon payments and principal payments. The bond portfolio can also generate losses from events such as defaults in the bonds and from profits or losses from trading activity. On the right side of the structure are the various classes of securities (or tranches) that provided the financing for the portfolio and that have claims of varying seniorities to receive cash inflows. A tranche is a security issued to finance a portfolio with specified claims to the cash flows from the portfolio. The cash flows from the collateral pool of assets are distributed using a waterfall approach. Without any defaults, the assets should generate $7 million in coupon income found as the product of the asset size ($100 million) and average coupon of the assets (7%). The first priority of the cash flows is to meet direct expenses and fees for the operation and management of the CDO. For simplicity, this example ignores these fees and expenses.

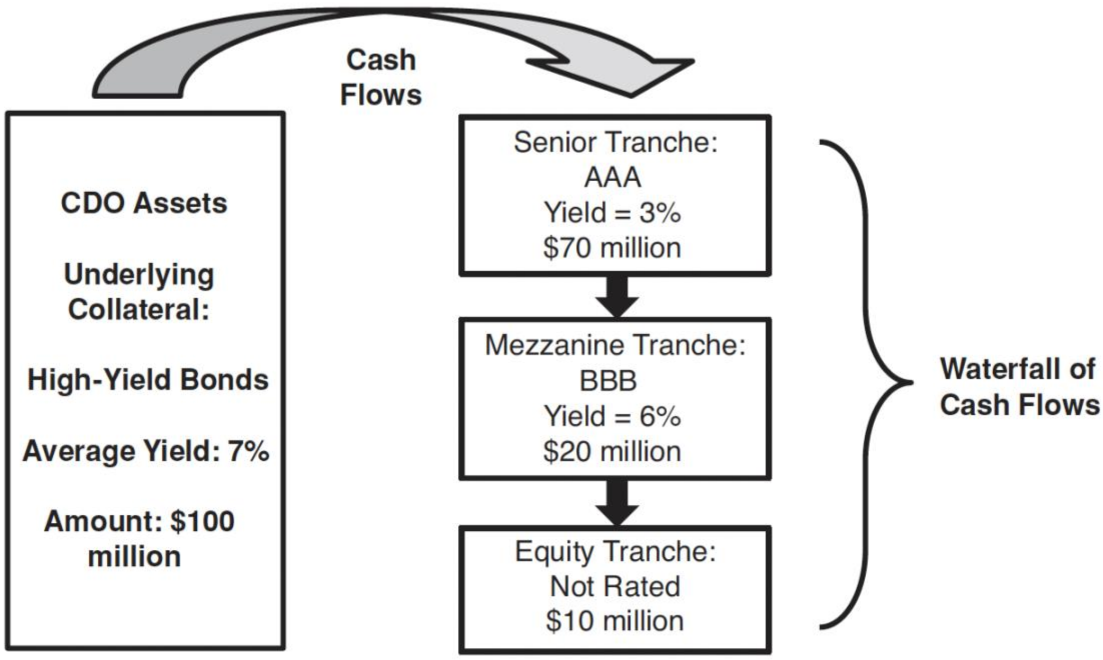


EXHIBIT 23.5 Simplified CDO Structure

After fees and expenses, the cash flows are distributed to the various tranches in order of their seniority. The senior tranche is owed $2.1 million per year in coupon income, found as the product of its size ($70 million) and coupon rate (3%). The mezzanine tranche is owed \$1.2 million, and the equity tranche has rights as residual claimants to the remaining \$3.7 million.

Example: Suppose that the CDO depicted in the above exhibit alters its portfolio such that the average coupon on the assets is 6%. Ignoring defaults, fees, and expenses, how much annual income should be available to the equity tranche? The answer is that \$3.3 million would go to the senior and mezzanine tranches and \$2.7 million would be available for the equity tranche.

The previous numerical example ignored defaults in the bond portfolio. However, this CDO contains bonds subject to substantial credit risk since they are below investment grade (i.e., are BB or lower), and therefore it is reasonable to expect defaults. When collateral assets default, the order of the waterfall reverses, relative to the order for receiving cash flows, such that the lowest seniority tranches experience the losses first. The losses from asset defaults are posted against the lowest remaining tranche until that tranche is wiped out, at which point the losses are posted against the next lowest seniority tranche. For example, if \$11 million of assets completely defaulted (i.e., with no recovery), the equity tranche would be wiped out, and the notional principal of the mezzanine tranche would be reduced from \$20 million to \$19 million.

Note that defaults in the CDO's collateral portfolio reduce both the left and right sides of the exhibit. The assets are reduced in size and in annual income. The aggregated tranches are reduced in size by reducing the tranches, starting with the lowest seniority tranche or tranches. As the debt tranches are reduced in notional principal, their claim to coupon income is reduced. Thus, if the mezzanine tranche in the example is reduced from \$20 million to \$19 million, the annual coupons owed to the mezzanine tranche holders would fall from \$1.20 million to \$1.14 million.

Example: Suppose that the CDO depicted in the exhibit experiences defaults in \$35 million of the assets with no recovery. What will happen to the mezzanine tranche and senior tranche? The answer is that after the equity tranche is eliminated due to the first \$10 million in defaults, the mezzanine tranche is eliminated due to the next \$20 million in defaults. The remaining $5 million of defaults will bring down the notional value of the senior tranche from $70 million to $65 million.

Attachment and detachment points

Note that the 20% of the structure in the exhibit above is represented by the mezzanine debt tranche that lies between the 70% financed by senior debt and the 10% financed by the junior-most tranche (the equity tranche). As losses to the collateral pool are experienced due to defaults in the portfolio of bonds, the first 10% of the losses are applied against the equity tranche, and the last 70% are applied against the senior tranche. Thus the losses to the mezzanine tranche begin when 10% of the collateral assets have defaulted and end when 30% of the collateral assets have defaulted and the mezzanine tranche is eliminated.

The first percentage loss in the collateral pool that begins to cause reduction in a tranche is known as the lower attachment point or simply the attachment point. The higher percentage loss point at which the given tranche is completely wiped out is known as the upper attachment point or the detachment point. Thus, the mezzanine tranche in the simplified example has a lower attachment point of 10% and an upper attachment point or detachment point of 30%. Each tranche is often identified using these points such that the mezzanine tranche in the example might be described as being a 10%/30% mezzanine tranche.

Balance Sheet CDOs

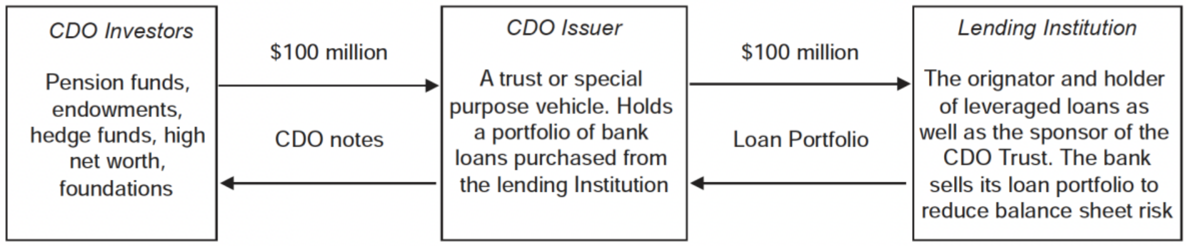
Banks and insurance companies are the primary sources of balance sheet CDOs. They use these structures to manage the assets on their balance sheets. In a balance sheet CDO, the seller of the assets, a financial institution, seeks to remove a portion of its loan portfolio or other assets from its balance sheet. The bank constructs an SPV to dispose of some of its balance sheet assets into the CDO structure. The CDO's asset manager is often the selling bank, where the bank is hired under a separate agreement to manage the portfolio of loans that it sold to the CDO trust. Also, the CDO trust will have a trustee whose job it is to protect the interests of the CDO tranche investors. Usually, this is not the bank or an affiliate because of conflict-of-interest provisions.

There are many potential conflicts of interest between commercial banking, securities services, and investment banking when they are all conducted under the same corporate umbrella. Some examples: When asked for advice by an investor, a bank might be tempted to recommend securities that the investment banking part of its organization is trying to sell. A bank, when it lends money to a company, often obtains confidential information about the company. It might be tempted to pass that information to the mergers and acquisitions arm of the investment bank to help it provide advice to one of its clients on potential takeover opportunities. The research end of the securities business might be tempted to recommend a company's share as a "buy" in order to please the company's management and obtain investment banking business.Suppose a commercial bank no longer wants a loan it has made to a company on its books because the confidential information it has obtained from the company leads it to believe that there is an increased chance of bankruptcy. It might be tempted to ask the investment bank to arrange a bond issue for the company, with the proceeds being used to pay off the loan. To avoid these, banks have are internal barriers known as *Chinese walls*. These internal barriers prohibit the transfer of information from one part of the bank to another when this is not in the best interests of one or more of the bank's customers.

Goals of a balance sheet CDO to the financial institution divesting the assets:

1. A bank or insurance company might wish to reduce its credit exposure to a particular client or industry, so it transfers these risks to the CDO;
2. The bank or insurance company might need a capital infusion; or
3. The bank or insurance company might wish to reduce its regulatory capital charges. By selling a portion of its loan or bond portfolio to a CDO, the institution can free up regulatory capital required to support those credit-risky assets.

Many balance sheet CDOs are self-liquidating. All interest and principal payments from the commercial loans are passed through to the CDO investors rather than being reinvested in new assets. Other balance sheet CDOs provide for the reinvestment of loan payments into additional commercial loans to be purchased by the CDO trust. After any reinvestment period, the CDO trust enters into an amortization period, when the loan proceeds are used to pay down the principal of the outstanding CDO tranches. The exhibit below shows the relations between CDO investors (which in this example put up $100 million in cash), the CDO issuer, and the lending institution.



Cash-funded CDOs vs Synthetic CDOs

In addition to balance sheet versus arbitrage CDOs, another major distinction between CDOs is that of cash-funded versus synthetic. The distinction focuses on whether the SPV obtains the risk of the portfolio using cash holdings or derivative positions. Synthetic balance sheet CDOs differ from the cash-funded variety in several important ways. First, cash funded CDOs are constructed with an actual sale and transfer of the loans or assets to the CDO trust. Ownership of the assets is transferred from the bank's balance sheet to that of the CDO trust in return for cash. In a synthetic CDO, however, the sponsoring bank or other institution transfers the risks and returns of a designated basket of loans or other assets via a credit derivative transaction, usually a credit default swap or a credit return swap. Therefore, the bank transfers its risk profile associated with its assets but does not give up the legal ownership of the assets and does not receive cash from selling assets.

Cash-funded CDOs and regulatory capital

A cash-funded CDO involves the actual purchase of the portfolio of securities serving as the collateral for the trust and to be held in the trust. In other words, physical ownership of the assets is acquired by the CDO. An advantage of a cash CDO is that it can be used to completely replace risky assets with cash on a financial institution's balance sheet, rather than synthetically removing only the risk through derivatives. There are several potential advantages to the financial institution from using a cash-funded balance sheet CDO. Banks are required by regulators to maintain a particular level of capital, depending on the risks of its assets. Higher-risk assets require higher quantities of regulatory capital. Banks maintain regulatory capital by obtaining financing through common stock and other sources of financing that are considered to be more expensive than sources of capital that do not serve as regulatory capital, such as deposits. Reducing risk-based/regulatory capital is the single most important motivation for a bank to form a CDO trust. Most major banks are required to maintain risk-based capital such as 8% of the outstanding balance of commercial loans. Using a CDO trust to securitize and sell a portfolio of commercial loans can free up regulatory capital that must be committed to support the loan portfolio.

Basel III is an internationally agreed set of measures developed by the Basel Committee on Banking Supervision in response to the financial crisis of 2007-09. The measures aim to strengthen the regulation, supervision and risk management of banks. Like all Basel Committee standards, Basel III standards are minimum requirements which apply to internationally active banks.On 1 January 2013 South Africa implemented amended Regulations which, in line with the Basel III framework, essentially address both bank-specific and broader, systemic risks. In South Africa, the branch capital of a bank must at all times be at least the greater of ZAR250 million or 8% of the amount of assets and other risk exposures of the branch. The branch must maintain a minimum reserve balance in an account with the SARB.

Example:

Consider a bank with a $500 million loan portfolio that it wishes to sell. It must hold risk-based capital equal to 8% × $500 million = $40 million to support these loans. If the bank sponsors a CDO trust where the trust purchases the $500 million loan portfolio from the bank and finds outside investors to purchase all of the CDO securities, the bank no longer has any exposure to the basket of commercial loans and now has freed $40 million of regulatory capital that it can use in other parts of its balance sheet. Sometimes the equity tranche of the CDO trust is unappealing to outside investors and cannot be sold. In this circumstance, the sponsoring bank may have to retain an equity or first-loss position in the CDO trust. If this is the case, the regulatory capital standards require the bank to maintain risk-based capital equal to its first-loss position. Thus, the bank needs to maintain $1 in regulatory capital for each $1 of ownership in an equity tranche.

Example:

Example: If the sponsoring bank from the previous example had to retain a $10 million equity piece in the CDO trust to attract other investors, it must take a one-for-one regulatory capital charge ($10 million) for this first-loss position. This means that only $30 million ($40 million minus $10 million) of regulatory capital is freed by the CDO trust. By selling existing loans into a CDO trust, a lending institution receives cash proceeds from the sale of its loans to the CDO trust that can be used to originate additional commercial loans or to strengthen its balance sheet. With its cash in hand, the bank can reduce its overall balance sheet by paying down its liabilities. Additionally, the selling bank may be able to reduce its credit exposure to one industry or group of borrowers if the bank deems that its exposures are too high. The bank can preserve customer relations by lending to a higher credit exposure to a particular client that it would otherwise wish in order to maintain its relationship with its borrower and then reduce its exposure through divesting some of the loans into a CDO.

Lifecycle of a CDO and basic overview

In most CDOs, there is a three-period life cycle. First, there is the *ramp-up period*, during which the CDO trust issues securities (tranches) and uses the proceeds from the CDO note sale to acquire the initial collateral pool (the assets). This is usually done over a period of one or two months. The CDOs' trust documents govern what type of assets may be purchased. The second phase is normally called the *revolving period*, during which the manager of the CDO trust may actively manage the collateral pool for the CDO, potentially buying and selling securities and reinvesting the excess cash flows received from the CDO collateral pool. This is usually done for a period of three to five years, but is dependent on the level of default sustained. The last phase is the *amortization period*. During this phase, the manager for the CDO stops reinvesting excess cash flows and begins to wind down the CDO by repaying the CDO debt securities. As the CDO collateral matures, the manager uses these proceeds to redeem the CDO's outstanding notes.

A major bank usually serves as the sponsor for the trust. The sponsor of the trust establishes the trust and bears the associated administrative and legal costs. At the center of every CDO structure is a *special purpose vehicle (SPV*), a legal entity established to accomplish a specific transaction, such as serving as the heart of a CDO structure. The SPV owns the collateral placed in the trust and issues notes and equity (tranches) against the collateral it owns. Often SPVs are referred to as *bankruptcy remote*. Bankruptcy remote means that if the sponsoring bank or money manager goes bankrupt, the CDO trust is not affected. In other words, the trust assets remain secure from any financial difficulties suffered by the sponsoring entity so that investors in the CDO tranches have a direct claim on the collateral. The CDOs are usually issued in different classes of securities (*tranches*). Each tranche of a CDO structure may have its own credit rating. Typically, most of the tranches of notes issued by the CDO receive an investment-grade rating by a nationally recognized statistical rating organization (NRSRO), with the exception of highly subordinated fixed-income tranches or the equity tranche. The equity tranche is the first-loss tranche. It is the last to receive any cash flows from the CDO collateral and the first tranche on the hook for any defaults or lost value of the CDO collateral. Often the equity tranche is held by the issuer of the trust.

The underlying portfolio or pool of assets (and/or derivatives) held in the SPV within the CDO structure is also known as the *reference portfolio*. Another term for the portfolio or the pool is the collateral.

Arbitrage CDOs

Arbitrage CDOs: The goal of these CDOs is to earn a positive spread between the cost of funds (the interest paid to CDO investors) to acquire the asset collateral and the return on the collateral. In other words, the CDO issuer seeks to generate a return from the assets that is greater than the rate paid to investors.

Arbitrage CDOs are primarily motivated by a goal of successful selection and management of the CDO’s collateral pool. A sponsor such as a money management firm establishes a CDO and takes an equity stake to earn a direct profit from the CDO. Arbitrage CDOs are designed to make a profit by capturing a spread for the equity investors in the CDO and from fees. The spread is captured as the excess of the higher-yielding securities that the CDO contains in its collateral portfolio and the yield that it must pay out on its fixed-income tranches issued to the CDO investors. Put differently, an arbitrage profit is earned if the CDO trust can issue its securities tranches at a yield substantially lower than the yield earned on the bond collateral contained in the trust, such that the equity tranche of the trust receives expected residual income disproportionate to its risk. Furthermore, money management firms earn fees on the amount of assets under management. By creating an arbitrage CDO, an investment management firm can increase its assets under management and therefore its income.

Another way to view the profit motive of an arbitrage CDO is in terms of market values rather than spreads and yields. The profit is earned by selling (issuing) securities (tranches) to outside investors at an aggregated price that is higher than that paid for all of the assets placed into the CDO structure as collateral. Thus the value of the equity tranche to the issuer could be greater than the money that the sponsor invested in the equity tranche.

Credit Risk Enhancements

Credit enhancement is a strategy for improving the credit risk profile of a business, usually to obtain better terms for repaying debt. In the financial industry, credit enhancement may be used to reduce the risks to investors of certain structured financial products.

Subordination is the most common form of credit enhancement in a CDO transaction, and it flows from the structure of the CDO trust. It is an internal credit enhancement. Subordination is the process of protecting a given security by issuing other securities that have a lower seniority to cash flows.

For instance, CDO trusts typically issue several classes or tranches of securities. The lower-level, or subordinated, tranches provide credit support for the higher-rated tranches. The equity tranche in a CDO trust is the first-loss position and therefore provides credit enhancement for every class of CDO securities above it. Junior tranches of a CDO are rated lower than the senior tranches but in return receive a higher coupon rate commensurate with their subordinated status and therefore greater credit risk.

The CDO structures can also be used for collateral assets with little or no credit risk, such as insured mortgages. In that case, subordination affects the timing of payments rather than the credit risk of payments to the various tranches. In a traditional sequential pay CDO, the principal of the senior tranches must be paid in full before any principal is paid to the junior tranches. This sequential payment structure is often referred to as the waterfall. As interest and principal payments are received from the underlying collateral, they flow down the waterfall: first to the senior tranches of the CDO trust and then to the lower-rated tranches. Subordinated tranches must wait for sufficient interest and principal payments to flow down the tranche structure before they can receive a payment.

Overcollateralization refers to the excess of assets over a given liability or group of liabilities. Overcollateralization results from the subordinated tranches in a CDO. The level of overcollateralization is the ratio of the assets available to meet an obligation to the size of the obligation and all other obligations senior to that obligation.